Abdelrahman Khalifa

+20 114 250 5042 | Cairo, Egypt | abdelrahman.khalifa.gad@gmail.com | linkedin.com/in/abd002 | github.com/Abd002

Summary

Software engineer with a strong algorithms background: Codeforces Expert, 4-time ECPC finalist and mentor in the local ICPC community. Built a Node.js/MongoDB REST backend and a Python app at EKSON, and got 11 PRs merged upstream as a Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting).

Experience

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Built the backend for remote operation and firmware updates: a Node.js/Express REST API with MongoDB that the desktop app polls to relay commands, store session and telemetry data, and deliver new firmware; GitHub Actions CI.
- Developed the Python (PyQt5) control app; used asyncio to wait for device and server messages concurrently, cutting CPU use and latency by about 40% (measured).

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Designed and built the **Printers page for COSMIC Settings** as 5 Rust crates, with the backend as a separate service behind a varlink API so a crash cannot take down Settings (cosmic-settings #2182, in review).
- **11 PRs merged upstream** (list): ported cups-rs to libcups3 with CI across architectures, added API bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 (#86977); Boa: faster `JsStr::code_points()` for Latin1 strings (PR #4553); Rust Clippy: fixed `ref_as_ptr` false positives (PR #16214).

Projects

Inventory Management System | *C#, SQL Server* GitHub

- Stock, orders and deliveries with role-based access; audit log with SQL Server triggers that record the logged-in user.

Sports Management Web App | *team; ASP.NET Core MVC, MS SQL, Entity Framework* GitHub

- Designed the relational schema and wrote the backend and account-management logic for role-based dashboards.

Database Systems, CMU 15-445 | *C++17, SQL* GitHub

- Thread-safe Count-Min Sketch for the BusTub database (per-row locks, merge, top-k); SQL analytics in DuckDB.

Technical Skills

Languages: Python, JavaScript (Node.js), SQL, C++, C, Rust, C#, Java
Backend & data: REST API design, Express, MongoDB, SQL Server, DuckDB, data modeling, ASP.NET Core MVC, Entity Framework
Computer science: Data structures and algorithms, complexity and optimization, concurrency, operating systems, database internals
Tools: Git/GitHub, GitHub Actions CI, pytest, Linux, debugging, code review

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors (84.6%).
- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024.